2017 HW2

Slew, Elmore Delay, and Gate Capacitance

EE3032 / Introduction to VLSI Design

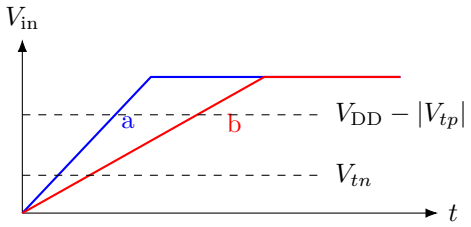
The supplied two-page handwritten answer was read first. All four original questions are answered below, with independently derived assumptions and an executable calculator. The circuit diagrams and calculations are independently reproducible.

1. Faster input rise makes output fall faster

An initially low input turns the pMOS on and the nMOS off, so $V_{\text{out}} \simeq V_{\text{DD}}$. As the input rises, the nMOS begins discharging C_L when $V_{\text{in}} > V_{tn}$, while the pMOS continues sourcing current until $V_{\text{in}} \gtrsim V_{\text{DD}} - |V_{tp}|$.

$$C_L \frac{dV_{\text{out}}}{dt} = I_p(V_{\text{in}}, V_{\text{out}}) - I_n(V_{\text{in}}, V_{\text{out}}).$$

The faster stimulus a brings the nMOS to strong drive sooner and cuts off the opposing pMOS sooner. The net discharge current $I_n - I_p$ becomes large earlier and the output traverses its 90%–10% interval more quickly. Stimulus b keeps both devices partly on for longer and initially gives a smaller nMOS overdrive. Thus its output has a longer fall time at the same load and transistor sizes. This is a finite-input-slew effect, not a change in capacitance.



Reference comparison. The supplied answer correctly relates the quicker threshold crossings of a to faster discharge. We retain that explanation and make the opposing pull-up current explicit.

2. NAND followed by an inverter

Use the prompt's effective on-resistance model and only the given total node capacitances C_F, C_Z. The internal NAND stack-node capacitance is not specified and is omitted in this two-node model. For the NAND output $F = \overline{AB}$, the worst pull-down traverses two nMOS in series, while the worst pull-up traverses only one pMOS:

$$\boxed{\tau_{F,HL} = 2R_nC_F}, \qquad \boxed{\tau_{F,LH} = R_pC_F}.$$

For final output $Z = AB$, add the first-stage contribution and the inverter-stage contribution:

$$\boxed{\tau_{Z,LH} = 2R_nC_F + R_pC_Z}, \qquad \boxed{\tau_{Z,HL} = R_pC_F + R_nC_Z}.$$

These are Elmore first-moment/time-constant estimates. If delay means an approximate 50% crossing, use $t_{pd} \simeq \ln(2)\tau \simeq 0.693\tau$ consistently. A multistage nonlinear CMOS waveform is not a single exact exponential.

The worst rising F transition is 11→01 or 11→10, with one active pMOS; 11→00 turns both pMOS on in parallel and has a smaller effective $R_p/2$. F falling corresponds to an input transition into 11. Z changes in the opposite direction to F. For a declared numerical example $R_n = 1\,\mathrm{k\Omega}$, $R_p = 2\,\mathrm{k\Omega}$, $C_F = 10$ fF, and $C_Z = 5$ fF:

Node/direction	Elmore τ (ps)	Approx. 50% delay (ps)
F, H–L	20.000	13.863
F, L–H	20.000	13.863
Z, H–L	25.000	17.329
Z, L–H	30.000	20.794

Reference comparison. The supplied answer introduces an additional stack-node capacitance C_1 and adds R_nC_1 in the NAND discharge term. That extension can be useful when C_1 is known, but the original prompt supplies only C_F, C_Z. We give the requested two-node expressions and state that internal-node parasitics require further data.

3. Mobility ratio at the inverter midpoint

At $V_{\mathrm{in}} = V_{\mathrm{out}} = V_{\mathrm{DD}}/2$, both devices are saturated for positive thresholds. Current equality gives

$$\frac{\mu_n C_{\mathrm{ox}}}{2}\,2\left(\frac{V_{\mathrm{DD}}}{2} - V_{tn}\right)^2 = \frac{\mu_p C_{\mathrm{ox}}}{2}\,4\left(\frac{V_{\mathrm{DD}}}{2} - |V_{tp}|\right)^2.$$

Therefore

$$\frac{\mu_n}{\mu_p} = 2\left(\frac{V_{\mathrm{DD}}/2 - |V_{tp}|}{V_{\mathrm{DD}}/2 - V_{tn}}\right)^2, \qquad \boxed{\mu_n/\mu_p = 2 \text{ if } V_{tn} = |V_{tp}|.}$$

The value 2 is the intended matched-threshold square-law result and agrees with the reference. If threshold magnitudes differ, the midpoint condition alone does not determine a unique mobility ratio; the general expression above applies.

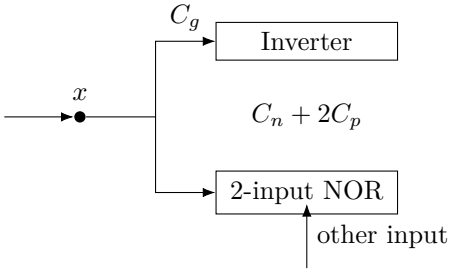
4. Capacitance at the fanout node x

Ignoring wire/drain capacitance, node x drives one inverter input and one NOR input. Let the inverter gate capacitances be C_n, C_p, with $C_g = C_n + C_p$. The NOR must have the same worst-case nMOS pull-down strength and equivalent pMOS pull-up strength as the inverter. Thus each NOR nMOS has the inverter nMOS width, and each of its two series pMOS has twice the inverter pMOS width (same L).

$$C_{\mathrm{NOR,pin}} = C_n + 2C_p, \qquad \boxed{C_x = C_g + C_{\mathrm{NOR,pin}} = 2C_n + 3C_p = 2C_g + C_p.}$$

For a balanced inverter $\beta_n = \beta_p$ with $\mu_n = 2\mu_p$, one needs $W_p = 2W_n$. Since gate capacitance is proportional to WL, $C_p = 2C_n$, giving

$$C_n = \frac{1}{3}C_g, \quad C_p = \frac{2}{3}C_g, \qquad \boxed{C_x = \frac{8}{3}C_g}.$$



Assumption check. The wording names β_p, β_n separately; it does not explicitly equate them. With equal lengths but an unbalanced inverter, the general result is

$$\frac{C_x}{C_g} = \frac{2\beta_n + 6\beta_p}{\beta_n + 2\beta_p} \quad \text{for } \mu_n = 2\mu_p.$$

Consequently 8/3 assumes the intended balanced reference inverter. The supplied answer uses that assumption and reaches the same result. The equal-equivalent-strength conditions for the NOR alone do not determine β_p/β_n. **Reproduce.** `python verify_design.py` evaluates every expression with declared parameters, exports `verification.json`, and checks the matched-threshold and balanced-inverter special cases. Use `-rn`, `-rp`, `-cf-ff`, `-cz-ff`, `-vtn`, `-vtp-abs`, and `-beta-p-over-n` to change the assumptions. These are analytical calculations, not fabricated HSPICE observations.